

Smart Food-Waste Reduction Framework

QSR Operations in South-East Nigeria · Technical Architecture & Master Plan

Executive Summary

This documentation details the complete architectural, algorithmic, and software blueprint for the **Smart Food-Waste Reduction Framework** tailored specifically to commercial Quick Service Restaurants (QSRs) operating across the South-East geo-political zone of Nigeria.

The system targets prominent chain brands including **Crunchies Fried Chicken, Kilimanjaro Restaurant, and Chicken Republic** across seven major South-East commercial centers: *Enugu, Awka, Owerri, Abakaliki, Umuahia, Onitsha, and Aba*.

Core Innovation

Eliminating kitchen overproduction by marrying pre-trained Time-Series Foundation Models (**Amazon Chronos-Bolt & Google TimesFM**) with a **South-East Nigerian Local Feature Layer** (Igbo market days, civil service paydays, and church surges) into a staggered prep batching engine.

1. Research Objectives

The study addresses six core objectives, with primary computational modeling focused on Objectives 3 & 4:

OBJECTIVE 1

Quantification & Characterization

Quantify edible food waste across operational stages: preparation, storage/spoilage, overproduction/buffet, and plate waste.

OBJECTIVE 2

Determinants Identification

Identify operational, technological, behavioral, and economic determinants of waste in the study area.

OBJECTIVE 3 (AI MODELING)

Effects of Smart Tech & Forecasting

Examine the mathematical effects of smart technologies, demand forecasting, inventory management, and menu engineering.

OBJECTIVE 4 (SOFTWARE SYSTEM)

Integrated Framework Development

Develop an integrated smart food-waste reduction platform tailored to QSR kitchen workflows in South-East Nigeria.

OBJECTIVE 5

Quasi-Experimental Pilot

Implement and validate the framework's effectiveness via an intervention pilot comparing Treatment vs Control branches.

OBJECTIVE 6

Policy & Scaling Guidelines

Establish operational standard operating procedures (SOPs) and policy guidelines for scaling across Nigeria.

2. Phased Research Methodology

The study adopts a mixed-methods design blending empirical waste measurements, econometric surveys, and AI software engineering:

Phase	Core Activity	Target Deliverable
Phase 1	Direct physical waste weighing across shifts	Baseline volume (kg/day) and financial loss (¥/day)
Phase 2	Staff surveys and managerial qualitative interviews	Statistical determinants and behavioral root causes
Phase 3	Model formulation with Chronos-Bolt and TimesFM	Predictive algorithms & menu engineering matrix
Phase 4	Django REST Framework backend & Swagger UI	Functional backend with digital prep sheet API
Phase 5	Quasi-experimental field intervention (8-12 weeks)	Pre- vs. Post-intervention comparative waste reduction %
Phase 6	Synthesis of operational SOPs and policy briefs	Sector-wide scaling blueprint for hospitality boards

3. AI Demand Forecasting Engine

Instead of training data-hungry neural networks from scratch, the system integrates two state-of-the-art **Time-Series Foundation Models (TSFMs)** in a dual-model weighted ensemble:

Amazon Chronos-Bolt

Tokenized language-model approach (T5).
Excels at tracking erratic localized spikes, sudden volume swings, and daily non-linear fluctuations.

Google TimesFM

Decoder-only patch transformer trained on 100B+ points. Delivers remarkable baseline stability, seasonal rhythms, and robust trend extrapolation.

Operational Rule: Disagreement-Based Staggered Batching

- **High Agreement (Difference < 8%):** High confidence. Kitchen cooks the full quota in standard opening batches.
- **High Disagreement (Difference ≥ 8%):** High volatility detected. The system automatically enforces a **2-stage prep**: cook 65% for the lunch rush, and prepare the remaining 35% on-demand during the afternoon. This stops overproduction waste.

4. South-East Nigeria Local Feature Layer

Global foundation models are calibrated with four essential regional operational drivers:



Igbo Market Day Cycles

4-day rotation (**Eke, Oye, Afor, Nkwo**) causing foot-traffic surges in major commercial hubs like Onitsha Main Market and Aba.



Salary / Payday Windows

Pronounced dining surges during the 25th–31st of each month in administrative civil-service capitals (Enugu, Abakaliki, Awka).



Sunday Church Surges

Substantial lunch rushes between 12:30 PM and 3:30 PM on Sundays, alongside Saturday wedding catering peaks.



Rainy Season Disruptions

Afternoon downpours that depress walk-in customer flow while triggering sudden spikes in deliveries and takeout.

5. Database Schema Blueprint

The backend is powered by 13 relational tables mapping directly to the study's research objectives:

Table Name	Cluster	Purpose & Key Fields
RestaurantChain	Organization	Parent brand name, headquarters city (Crunchies, Kilimanjaro, etc.)
Branch	Organization	Name, city, state, standby generator flag, is_pilot_intervention (Treatment vs Control)
UserProfile	Accounts	User link, branch assignment, system roles (Cook, Storekeeper, Manager, Auditor)
Ingredient	Menu/Inventory	Raw item name, unit of measure, unit cost in Naira (₦), chilled and ambient shelf-life
MenuItem	Menu	Finished dish name, category (Rice, Protein, Sides), selling price in ₦
RecipeBOM	Menu	Bill of Materials: maps dish portions to raw ingredient quantities with shrinkage %
InventoryBatch	Inventory	Stock batch received, quantity remaining, expiry date (FEFO rotation order)
PowerOutageLog	Inventory	Records duration of electrical power/generator outages affecting refrigeration
LocalCalendarContext	Calendar	Date, city, traditional market day, payday window flag, weather condition
SalesRecord	Sales	POS transactions: branch, dish, sale date, shift, portions sold, gross revenue (₦)
DailyForecastItem	Forecasting	Chronos prediction, TimesFM prediction, ensemble forecast, batch recommendation

Table Name	Cluster	Purpose & Key Fields
KitchenPrepTarget	Forecasting	Calculated raw ingredients (kg) to prepare per shift, actual used, variance
WasteLog	Waste	Disposal stage, menu item or raw ingredient, weight (kg), auto-calculated W loss, root cause
MenuEngineeringSnapshot	Analytics	Dishes classified into Star, Plowhorse, Puzzle, Dog with operational recommendations

6. Interactive Swagger UI API Specification

All endpoints are documented with OpenAPI 3.0 via drf-spectacular and testable at `/api/docs/`:

Method	Endpoint	Functional Action
POST	<code>/api/v1/auth/login/</code>	Authenticate staff credentials and issue API token
GET	<code>/api/v1/organization/branches/</code>	List all QSR branches with city and pilot intervention tags
GET	<code>/api/v1/menu/items/</code>	Retrieve dishes with retail prices and Recipe BOM linkages
GET	<code>/api/v1/calendar/today/?city=ONITSHA</code>	Get today's Igbo market day, payday window, and weather
POST	<code>/api/v1/waste/logs/</code>	Log waste event (kg, stage, cause) with instant ₦ loss valuation
GET	<code>/api/v1/waste/daily-summary/</code>	Fetch shift waste totals and stage breakdown in Naira
POST	<code>/api/v1/forecast/trigger/</code>	Execute Chronos + TimesFM ensemble & generate kitchen prep targets
GET	<code>/api/v1/forecast/prep-sheet/</code>	Serve tomorrow's digital kitchen prep sheet to cooks
GET	<code>/api/v1/analytics/menu-engineering/</code>	Retrieve Star/Plowhorse/Puzzle/Dog menu profitability matrix
GET	<code>/api/v1/analytics/pilot-comparison/</code>	Compare Treatment vs Control branches for Phase 5 evaluation

7. Implementation Roadmap

Development & Verification Status

- ✓ **Milestone 1:** Django project initialization & Swagger UI setup (Completed)
- ✓ **Milestone 2:** All 9 application modules & database models implemented (Completed)
- 🕒 **Milestone 3:** Database migration execution and sample data seeding (Ready to run)
- 🕒 **Milestone 4:** Model inference integration with Chronos-Bolt and Google TimesFM
- 🕒 **Milestone 5:** Quasi-experimental pilot deployment across South-East Nigeria